問題 285

次の角を表す動径を図示せよ.

解答:

360° で割った余りで位置を判定.

(1) $630° = 360° + 270° \to 270°$ の位置（y 軸負方向）

(2) $-310° = 50°$（第 1 象限）

(3) $490° = 130°$（第 2 象限）

(4) $-1120° = -1120 + 4 \times 360 = 320°$（第 4 象限）

(5) $2000° = 2000 - 5 \times 360 = 200°$（第 3 象限）

問題 286

次の角は第何象限の角か.

解答:

(1) 210°：第 3　(2) 400° = 40°：第 1　(3) −740° = 340°：第 4

(4) 820° = 100°：第 2　(5) −635° = 85°：第 1

問題 287

次の三角関数の値を求めよ.

解答:

(1) $\sin 210° = -\sin 30° = -\dfrac{1}{2}$

(2) $\cos 570° = \cos 210° = -\dfrac{\sqrt{3}}{2}$

(3) $\tan 390° = \tan 30° = \dfrac{\sqrt{3}}{3}$

(4) $\sin 630° = \sin 270° = -1$

(5) $\cos(-225°) = \cos 225° = -\dfrac{\sqrt{2}}{2}$

(6) $\tan(-480°) = \tan 240° = \sqrt{3}$

問題 288

次の角を弧度法で表せ.

解答:

(1) $135° = \dfrac{3\pi}{4}$　(2) $36° = \dfrac{\pi}{5}$　(3) $-10° = -\dfrac{\pi}{18}$

(4) $240° = \dfrac{4\pi}{3}$　(5) $-190° = -\dfrac{19\pi}{18}$

問題 289

次の角を 60 分法で表せ.

解答:

(1) $\dfrac{\pi}{3} = 60°$　(2) $\dfrac{5\pi}{4} = 225°$　(3) $-\dfrac{2\pi}{5} = -72°$

(4) $\dfrac{7\pi}{3} = 420°$　(5) $-\dfrac{\pi}{9} = -20°$

問題 290

次の三角関数の値を求めよ.

解答:

(1) $\sin\dfrac{5\pi}{3} = -\sin\dfrac{\pi}{3} = -\dfrac{\sqrt{3}}{2}$

(2) $\cos\dfrac{5\pi}{4} = -\cos\dfrac{\pi}{4} = -\dfrac{\sqrt{2}}{2}$

(3) $\tan\dfrac{\pi}{6} = \dfrac{\sqrt{3}}{3}$

問題 291

弧の長さと扇形の面積を求めよ.

解答:

$l = r\theta,\ S = \dfrac{1}{2}r^2\theta$

(1) 半径 4，中心角 $\dfrac{\pi}{6}$：$l = \dfrac{2\pi}{3},\ S = \dfrac{4\pi}{3}$

(2) 半径 3，弧 2π：$\theta = \dfrac{2\pi}{3},\ S = 3\pi$

問題 292

次の等式を証明せよ.

解答:

(1) $\tan\theta + \dfrac{1}{\tan\theta} = \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \dfrac{1}{\sin\theta\cos\theta}$　□

(2) $\dfrac{1}{1-\cos\theta} + \dfrac{1}{1+\cos\theta} = \dfrac{2}{1-\cos^2\theta} = \dfrac{2}{\sin^2\theta}$　□

問題 293

象限と 1 つの三角関数の値から残りを求めよ.

解答:

(1) 第 3 象限，$\sin\theta = -\dfrac{4}{5}$

$\cos\theta = -\dfrac{3}{5},\ \tan\theta = \dfrac{4}{3}$

(2) 第 4 象限，$\cos\theta = \dfrac{1}{3}$

$\sin\theta = -\dfrac{2\sqrt{2}}{3},\ \tan\theta = -2\sqrt{2}$

(3) 第 3 象限，$\tan\theta = 3$

$\cos\theta = -\dfrac{\sqrt{10}}{10},\ \sin\theta = -\dfrac{3\sqrt{10}}{10}$

問題 294

次の式を簡単にせよ.

解答:

(1) $\cos\theta\sin\left(\dfrac{\pi}{2}-\theta\right) + \sin\theta\cos\left(\dfrac{\pi}{2}-\theta\right) = \cos^2\theta + \sin^2\theta = 1$

(2) $\sin\left(\dfrac{\pi}{2}-\theta\right) - \sin(\pi+\theta) + \cos\left(\dfrac{\pi}{2}+\theta\right) + \cos(\pi-\theta)$

$= \cos\theta + \sin\theta - \sin\theta - \cos\theta = 0$

(3) $\tan(\pi+\theta)\sin\left(\dfrac{\pi}{2}+\theta\right) + \cos(\pi-\theta)\tan(\pi-\theta)$

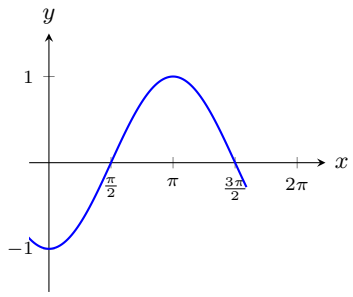
$= \tan\theta\cos\theta + (-\cos\theta)(-\tan\theta) = \sin\theta + \sin\theta = 2\sin\theta$

問題 295

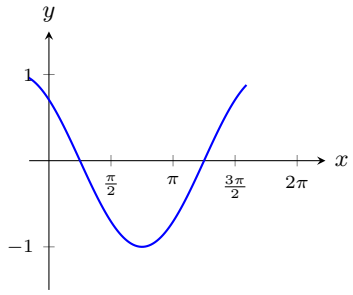
次の関数の周期を求め，グラフをかけ．

解答:

(1) $y = \sin\left(x - \dfrac{\pi}{2}\right) = -\cos x$　周期 2π



(2) $y = \cos\left(x + \dfrac{\pi}{4}\right)$　周期 2π

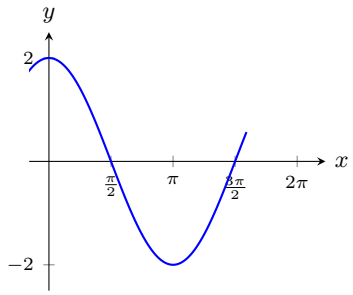


問題 296

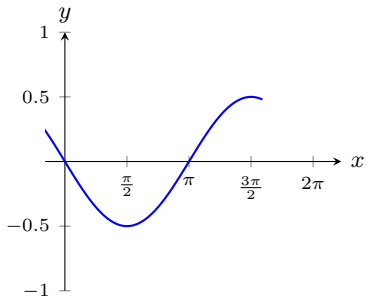
次の関数の周期を求め，グラフをかけ．

解答:

(1) $y = 2\cos x$　周期 2π，振幅 2



(2) $y = -\dfrac{1}{2}\sin x$　周期 2π，振幅 $\dfrac{1}{2}$



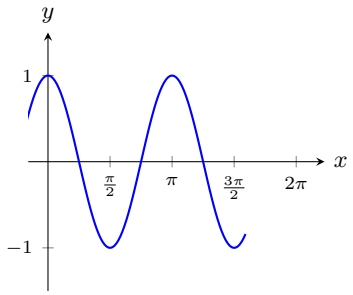
問題 297

次の関数の周期を求め，グラフをかけ．

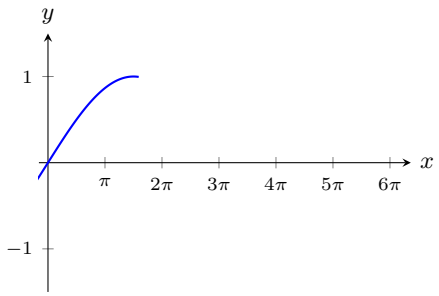
解答:

$y = \sin(kx)$ の周期は $\dfrac{2\pi}{|k|}$.

(1) $y = \cos 2x$　周期 π



(2) $y = \sin\dfrac{x}{3}$　周期 6π



問題 298

$0 \leqq x < 2\pi$ で方程式・不等式を解け．

解答:

(1) $\sin x = \dfrac{\sqrt{2}}{2} \to x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}$

(2) $\cos x = \dfrac{1}{2} \to x = \dfrac{\pi}{3}, \dfrac{5\pi}{3}$

(3) $\sin x \geqq \dfrac{\sqrt{3}}{2} \to \dfrac{\pi}{3} \leqq x \leqq \dfrac{2\pi}{3}$

(4) $\cos x < -\dfrac{\sqrt{2}}{2} \to \dfrac{3\pi}{4} < x < \dfrac{5\pi}{4}$

問題 299

$0 \leqq x < 2\pi$ で方程式を解け．

解答:

(1) $\tan x = 0 \to x = 0,\ \pi$

(2) $\tan x = -1 \to x = \dfrac{3\pi}{4},\ \dfrac{7\pi}{4}$